



Color Space

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Color Vision



What you see
(Red, Green, Blue)



What dogs & cats see
(Mostly Grey, some Blue and Yellow)

<http://www.colormatters.com/color-matters-for-kids/how-animals-see-color>

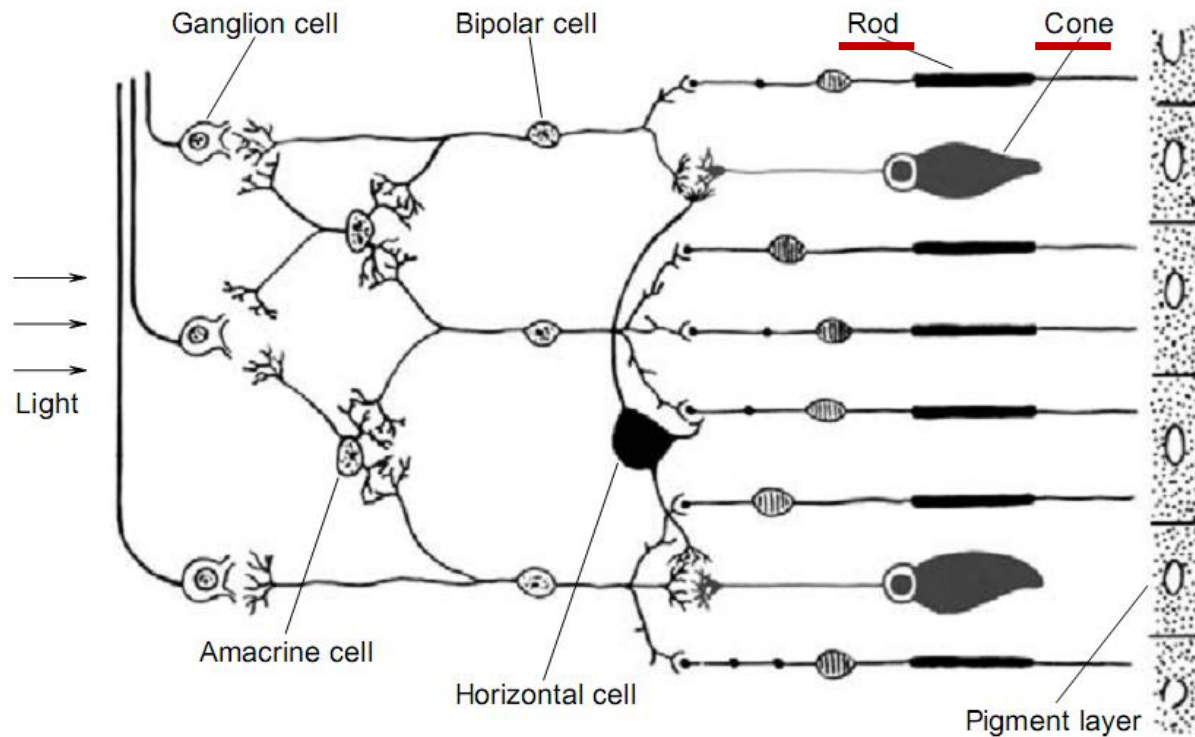
Photoreceptors

Rod

For scotopic vision at low light levels
(night/gray vision)
100 million rods in each eye

Cone

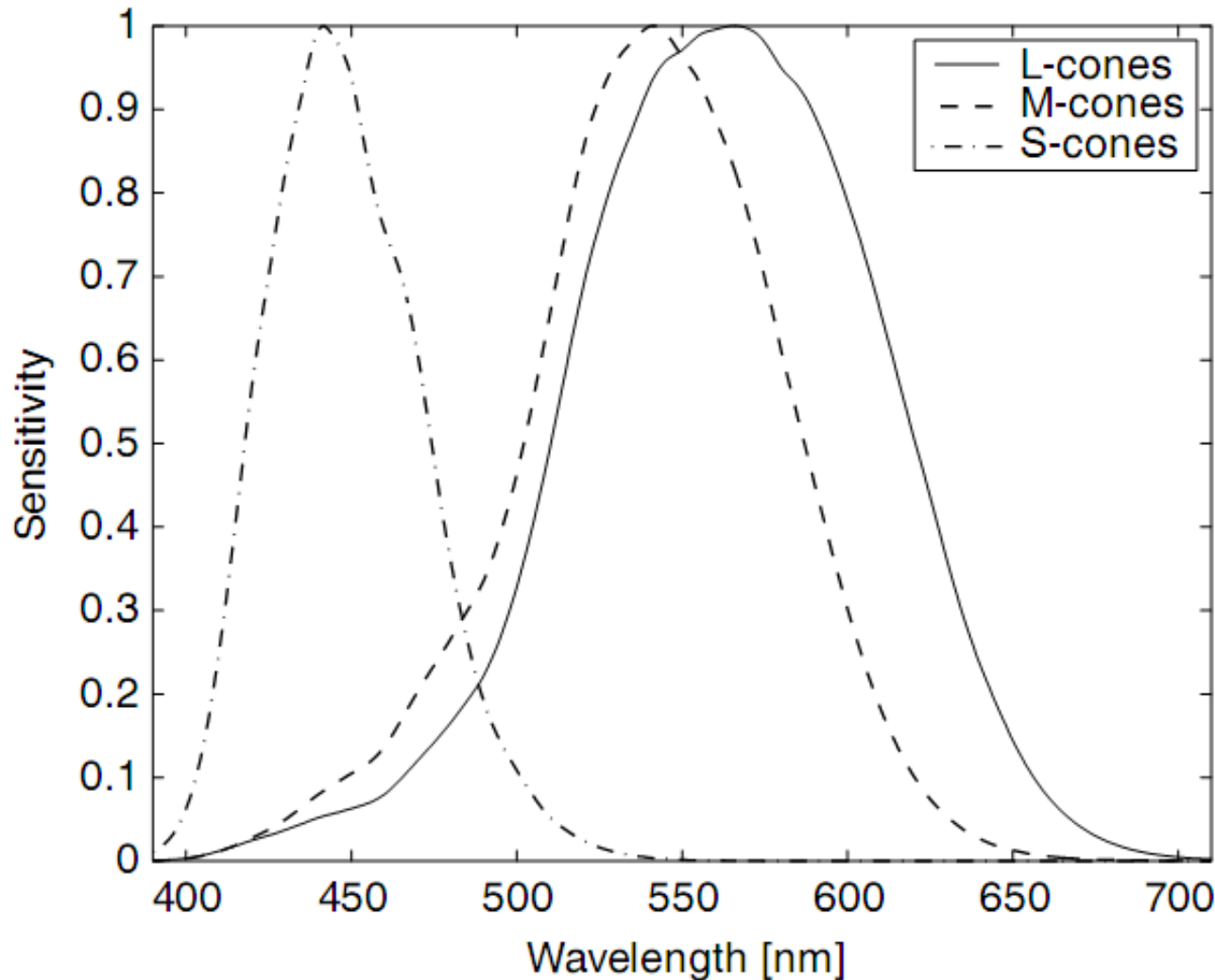
For photopic vision at high light levels
(daylight/color vision)
5 million cones in each eye



Anatomy of Retina

Ref: *Digital Video Quality*

Absorption Spectra of Three Cone Types



570nm (R)
540nm (G)
440nm (B)
(Peak)

L- and M-cones are similar, and S-cones are different.

Foundation of three primary colors

Ref: *Digital Video Quality*

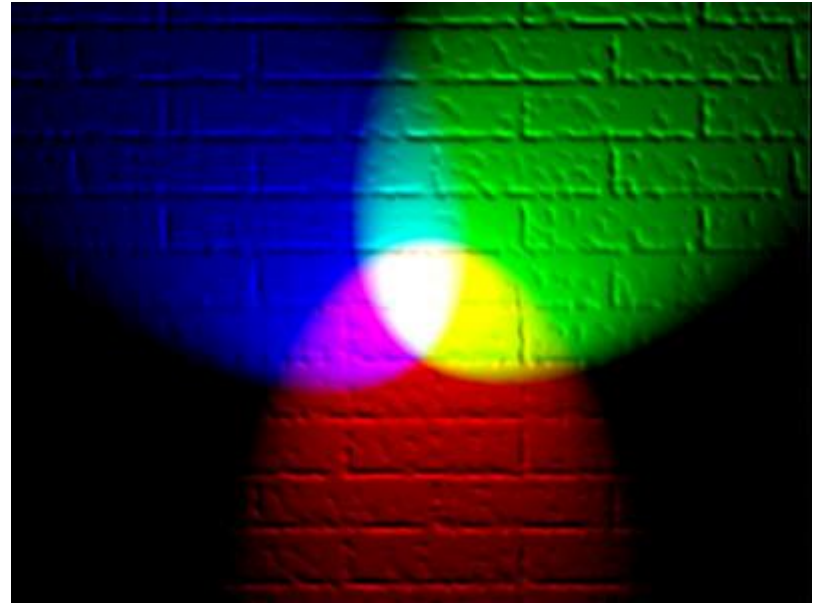
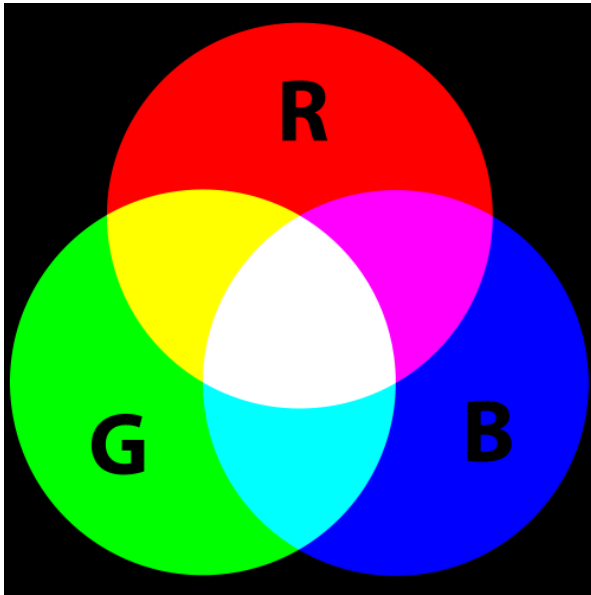


Color Space

- Additive Color Space for Display
- Chromaticity
- Color Space Transformation
- Color Specification

Other color models (not covered):
Subtractive color model (CMYK) for printing,
Human vision approximation model ($L^*a^*b^*$).

Additive Color



Additive color created by mixing light of colors, usually the three primary colors: red, green, and blue. (used in display devices)



Ref: Wikipedia

Tristimulus Color Reproduction

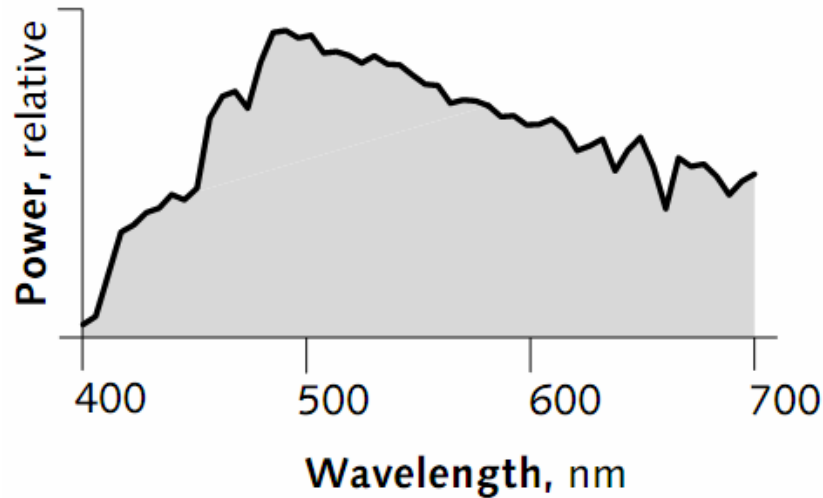
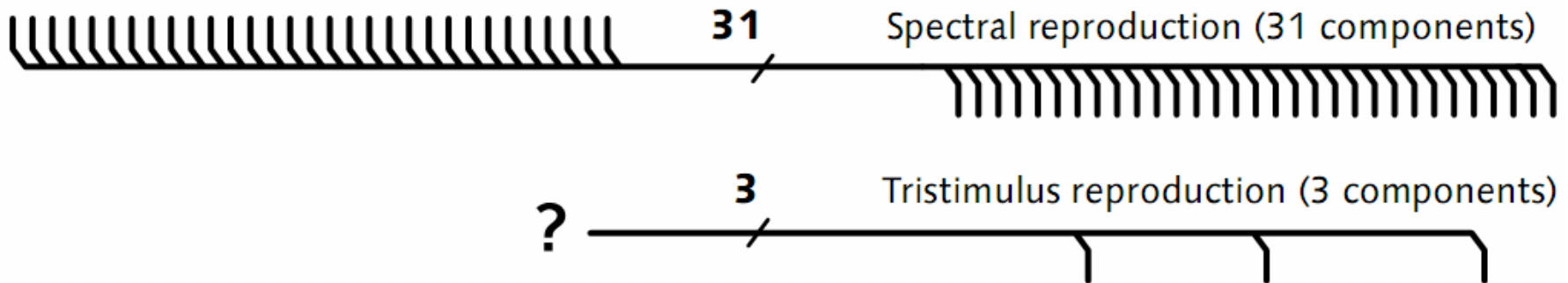
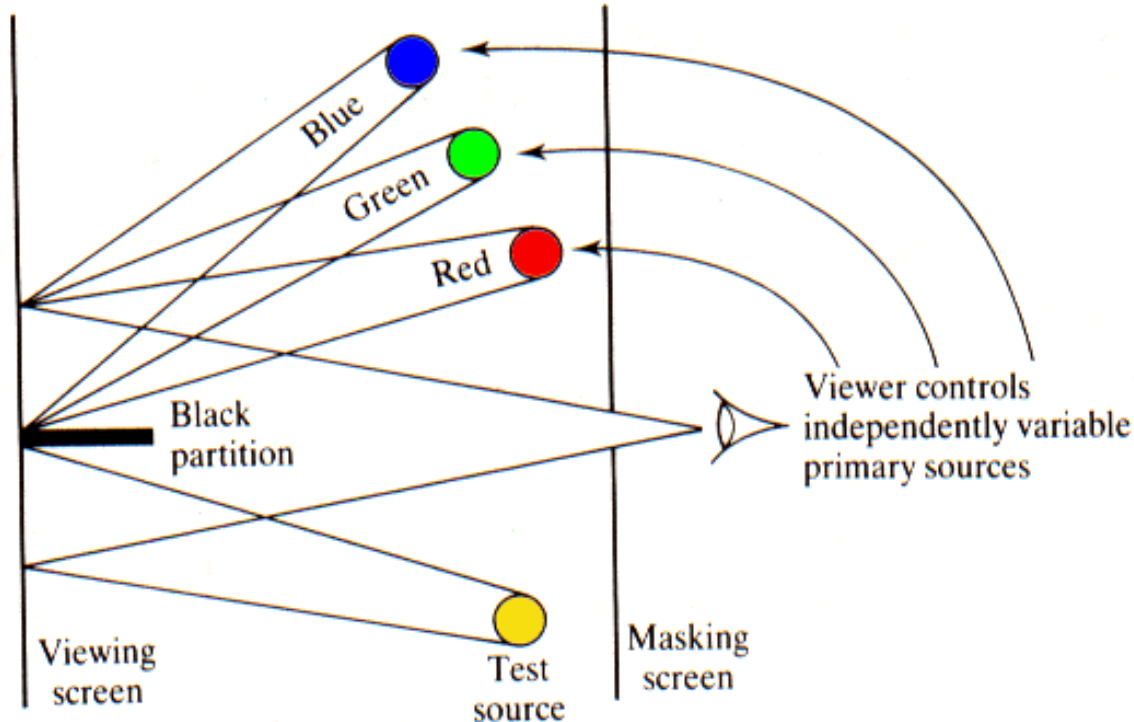


Figure 21.2 Spectral and tristimulus color reproduction. A color can be represented as a spectral power distribution (SPD), perhaps in 31 components representing power in 10 nm bands over the range 400 nm to 700 nm. However, owing to the trichromatic nature of human vision, if appropriate spectral weighting functions are used, three components suffice to represent color. The SPD shown here is the CIE D₆₅ daylight illuminant.



Tristimulus Color Matching

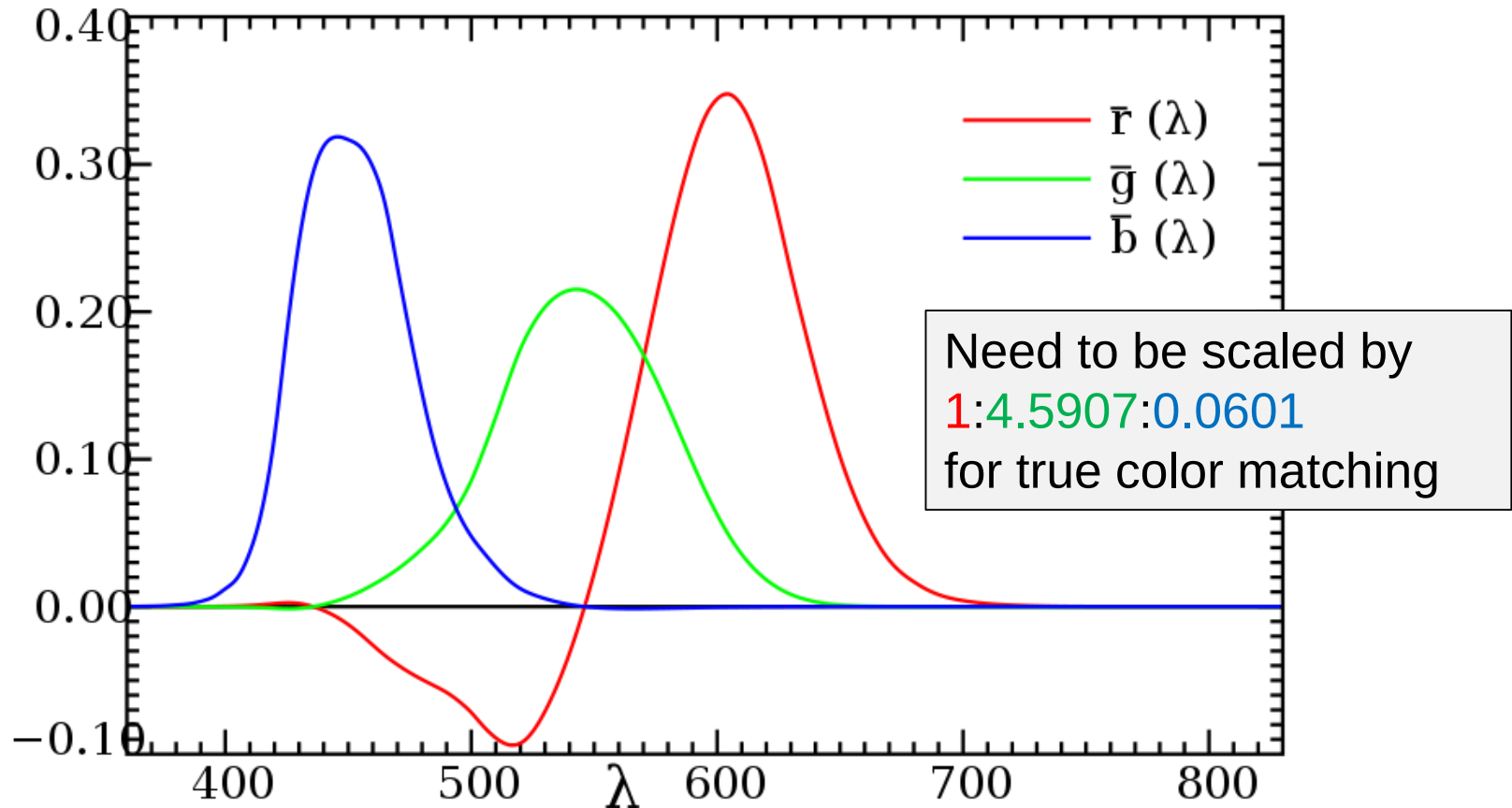
- Match the color using RGB primaries by adjusting the intensity weighting



Ref: <http://www.willamette.edu/~gorr/classes/GeneralGraphics/Color/tristimulus.htm>

CIE 1931 RGB Color Space

International Commission on Illumination (usually abbreviated **CIE** for its French name, Commission internationale de l'éclairage)



Color Matching Functions (CMF) in wavelength for standard observers using monochromatic lights **700nm**, **546.1nm**, and **435.8nm**.



Grassmann's Law

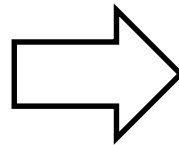
- The SPDs of the same color map to (via color matching) the same RGB
- Linearity of SPDs (mixture of colors) preserves in color space

$$R = \int I(\lambda) \bar{r}(\lambda) d\lambda$$

$$G = \int I(\lambda) \bar{g}(\lambda) d\lambda$$

$$B = \int I(\lambda) \bar{b}(\lambda) d\lambda$$

Linearity



$$I(\lambda) = I_1(\lambda) + I_2(\lambda)$$

$\Rightarrow$

$$(R, G, B) = (R_1, G_1, B_1) + (R_2, G_2, B_2)$$

Every color's SPD $I(\lambda)$
has its matched (R,G,B)

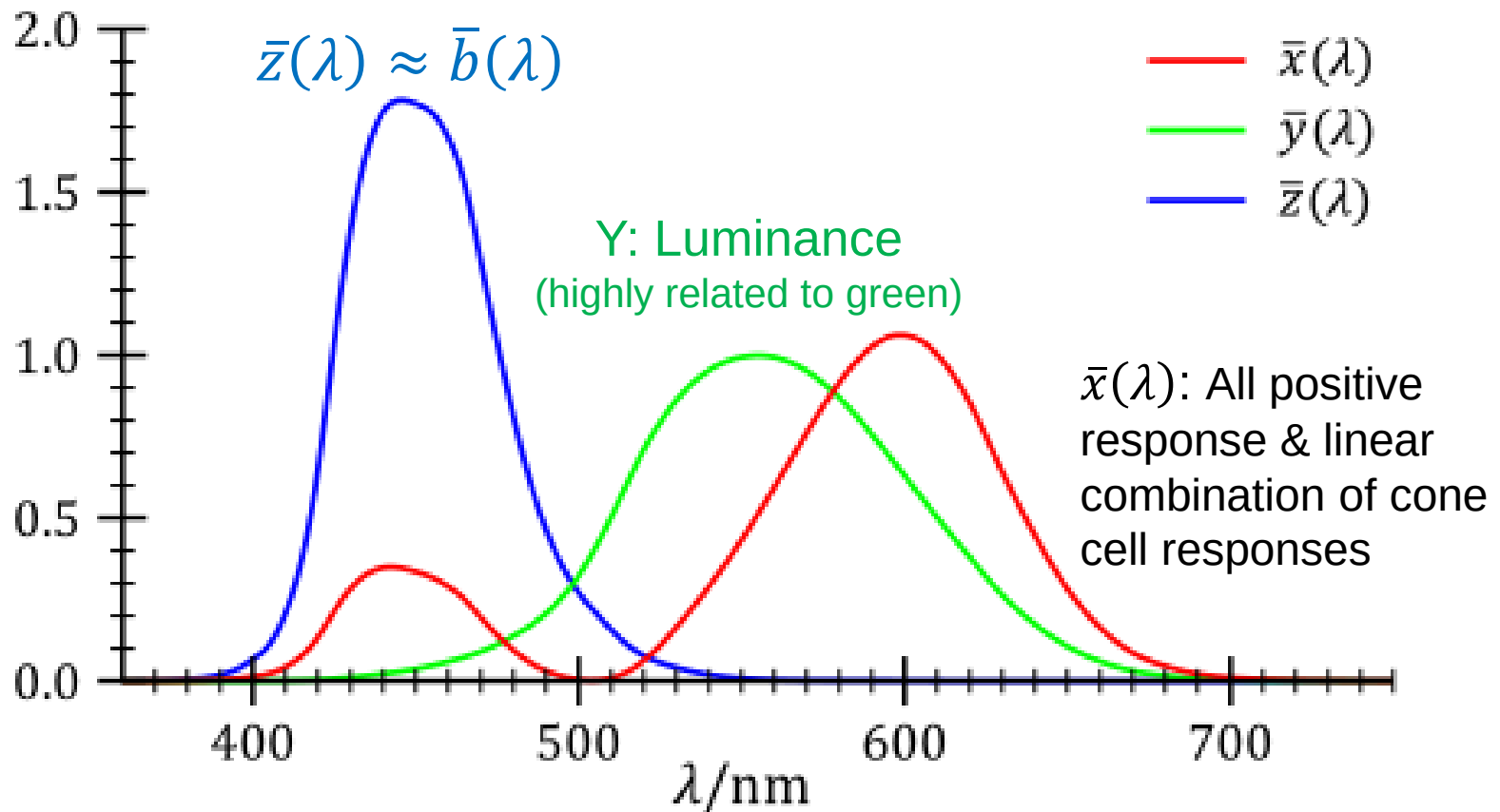
The color space for any given
primaries is the corresponding
linear space



Note on Tristimulus Color Space

- Not any three colors can be the primaries. They should be covered by L-/M-/S-cone's response for human vision.
 - Is CIE 1931 RGB sufficient?
 1. It has negative components, which will make the color space outside the color triangle.
 2. People want to have the Luminance as one of the primary.
- ➡ Generate another XYZ space for color space reference, but the primaries don't have real color (unable to reproduce colors directly).

CIE 1931 XYZ Color Space



For “interpretation” of color matching (device-independent),
but not for “reproduction” (device-dependent)

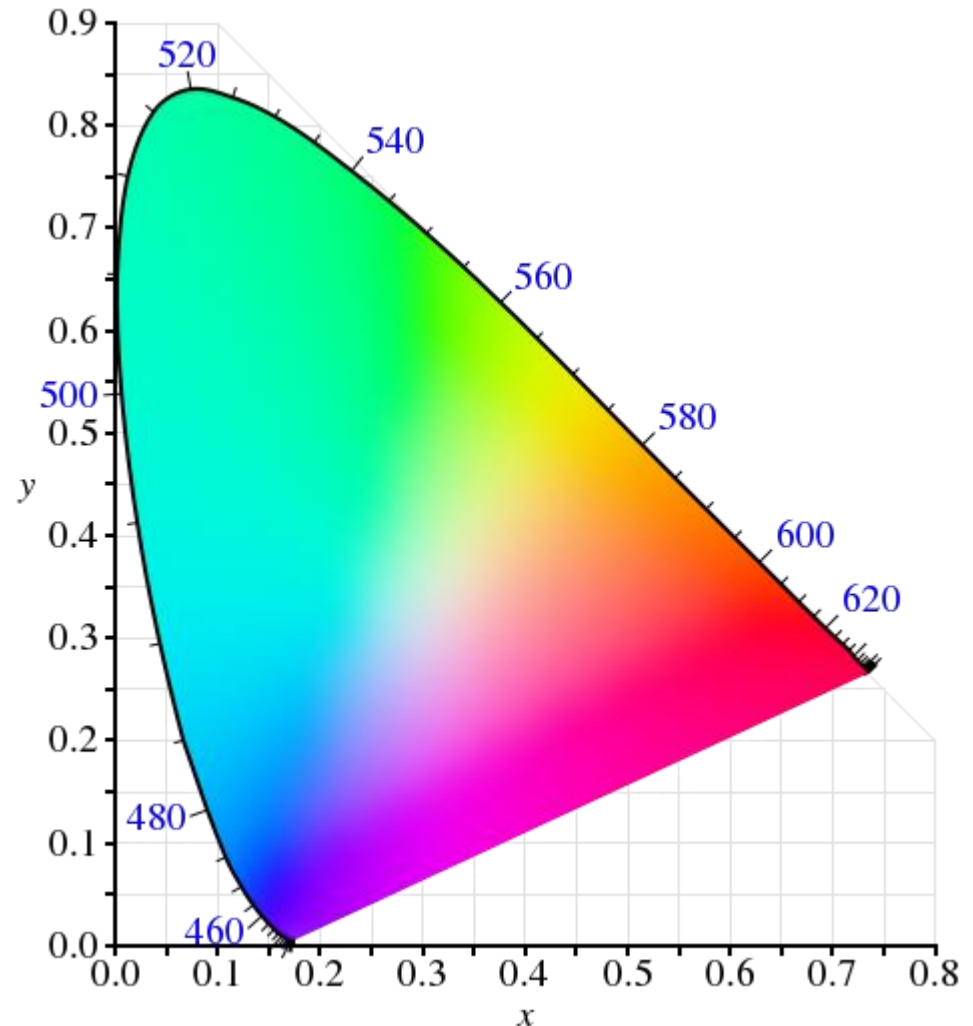
Ref: Wikipedia

CIE [x,y] Chromaticity

- “pure” color in the absence of lightness

$$x = \frac{X}{X + Y + Z}$$

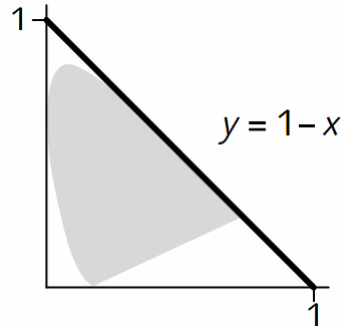
$$y = \frac{Y}{X + Y + Z}$$



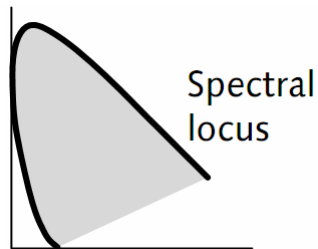
Ref: Wikipedia

CIE [x,y] Chart Features

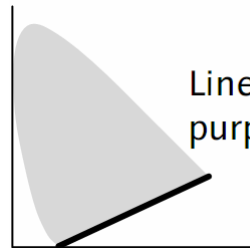
No blue
($z=0$)



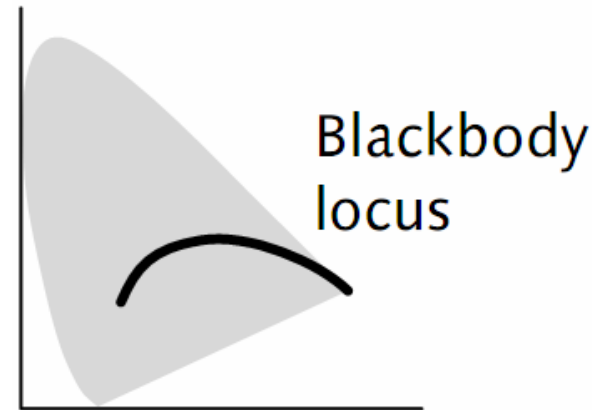
Monochromatic
light



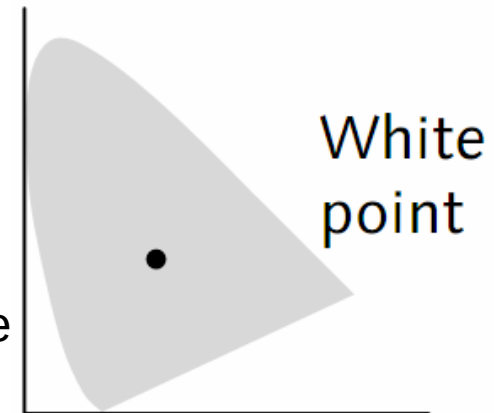
Line of
purples



Blackbody
radiator's
SPD (or color)
dependent on
temperature

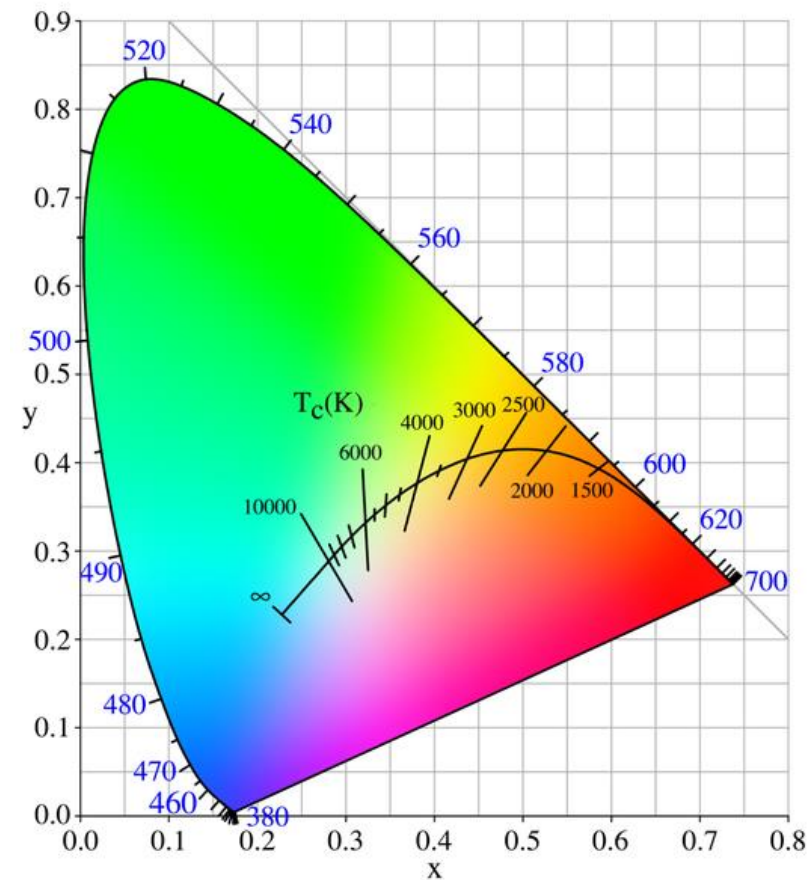
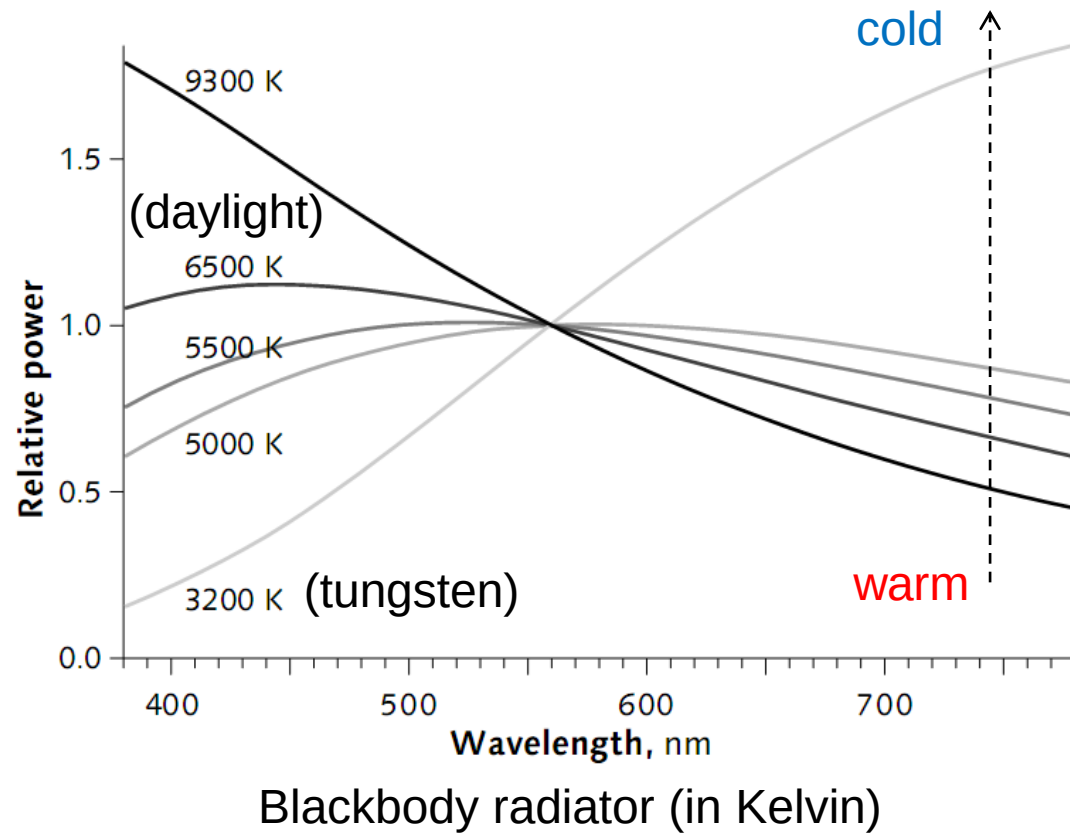


White is not
unique, and
depends on
color temperature



Color Temperature

Define the illuminant



Ref: Digital Video and HDTV, Wikipedia



“White”

- Given white reference SPD, “white” is defined by the matched chromaticity coordinates
 - The color appears on white object (uniform reflectance)

<i>Notation</i>	<i>x</i>	<i>y</i>	<i>z</i>
CIE III. A (obsolete)	0.4476	0.4074	0.1450
CIE III. B (obsolete)	0.3484	0.3516	0.3000
CIE III. C (obsolete)	0.3101	0.3162	0.3737
CIE III. D ₅₀	0.3457	0.3587	0.2956
CIE III. D ₅₅	0.3325	0.3476	0.3199
CIE III. D ₆₅ (daylight, 6500K)	0.312727	0.329024	0.358250
CIE III. E (equi-energy)(pure white)	0.333334	0.333330	0.333336
9300 K (discouraged, but used in studio standards in Japan)	0.283	0.298	0.419

Additive Reproduction using RGB primaries

- e.g. Rec. 709/sRGB primaries

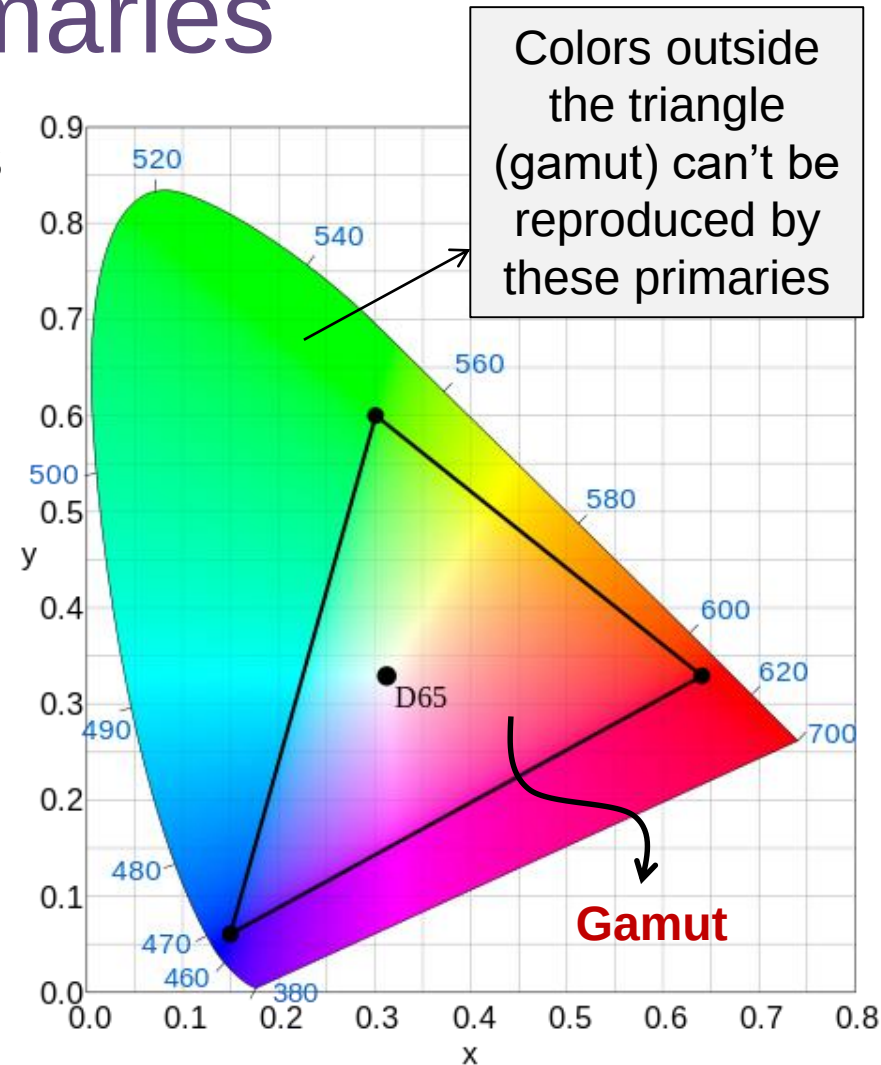
	Red	Green	Blue	White, D_{65}
x	0.640	0.300	0.150	0.3127
y	0.330	0.600	0.060	0.3290
z	0.030	0.100	0.790	0.3582

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = T \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Transformation
between RGB and
XYZ

$$\begin{bmatrix} R_d \\ G_d \\ B_d \end{bmatrix} = T_d^{-1} \cdot T_s \cdot \begin{bmatrix} R_s \\ G_s \\ B_s \end{bmatrix}$$

Transformation
between RGB
systems



Ref: Digital Video and HDTV, Wikipedia



Compute the Matrix T

- Use specified chromaticity of

- R/G/B

- white point

- And assume

- [R=1 G=1 B=1] corresponds to white point with $Y_w=1$

- All zero for black point

$$\begin{matrix} \begin{bmatrix} X_r & X_g & X_b \\ Y_r & Y_g & Y_b \\ Z_r & Z_g & Z_b \end{bmatrix} & = & \begin{bmatrix} x_r & x_g & x_b \\ y_r & y_g & y_b \\ z_r & z_g & z_b \end{bmatrix} \begin{bmatrix} A_r & 0 & 0 \\ 0 & A_g & 0 \\ 0 & 0 & A_b \end{bmatrix} \\ \mathbf{T} & & \mathbf{C} & & \mathbf{A} \end{matrix}$$

$$\begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} = \begin{bmatrix} x_w/y_w \\ 1 \\ z_w/y_w \end{bmatrix} = T \cdot \begin{bmatrix} R=1 \\ G=1 \\ B=1 \end{bmatrix} = \begin{bmatrix} x_r & x_g & x_b \\ y_r & y_g & y_b \\ z_r & z_g & z_b \end{bmatrix} \begin{bmatrix} A_r \\ A_g \\ A_b \end{bmatrix} \Rightarrow \begin{bmatrix} A_r \\ A_g \\ A_b \end{bmatrix} = C^{-1} \begin{bmatrix} x_w/y_w \\ 1 \\ z_w/y_w \end{bmatrix}$$

Primaries Transformation

- Rec. 709 RGB
$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.412453 & 0.357580 & 0.180423 \\ 0.212671 & 0.715160 & 0.072169 \\ 0.019334 & 0.119193 & 0.950227 \end{bmatrix} \cdot \begin{bmatrix} R_{709} \\ G_{709} \\ B_{709} \end{bmatrix}$$

Luminance contribution

$$\begin{bmatrix} R_{709} \\ G_{709} \\ B_{709} \end{bmatrix} = \begin{bmatrix} 3.240479 & -1.537150 & -0.498535 \\ -0.969256 & 1.875992 & 0.041556 \\ 0.055648 & -0.204043 & 1.057311 \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

- SMPTE 240M => Rec. 709

$$\begin{bmatrix} R_{709} \\ G_{709} \\ B_{709} \end{bmatrix} = \begin{bmatrix} 0.939555 & 0.050173 & 0.010272 \\ 0.017775 & 0.965795 & 0.016430 \\ -0.001622 & -0.004371 & 1.005993 \end{bmatrix} \cdot \begin{bmatrix} R_{145} \\ G_{145} \\ B_{145} \end{bmatrix}$$

Camera White Balance



- White balance (WB)
 - Under different illuminants, white paper can be all perceived as white by human even if SPD differs.
 - WB achieves this by making white object looks white, instead of reproducing the perceived color.
- Camera white balance in linear RGB
 - Given white point's captured values (R_w, G_w, B_w)
 - $\Rightarrow$ WB: Multiply captured values by $(R_w^{-1}, G_w^{-1}, B_w^{-1})$, such that white point becomes $(1,1,1)$

Note: 1. Not primaries transformation (no x/y/z involved)

2. $(R_w^{-1}, G_w^{-1}, B_w^{-1})$ similar with (A_r, A_g, A_b)

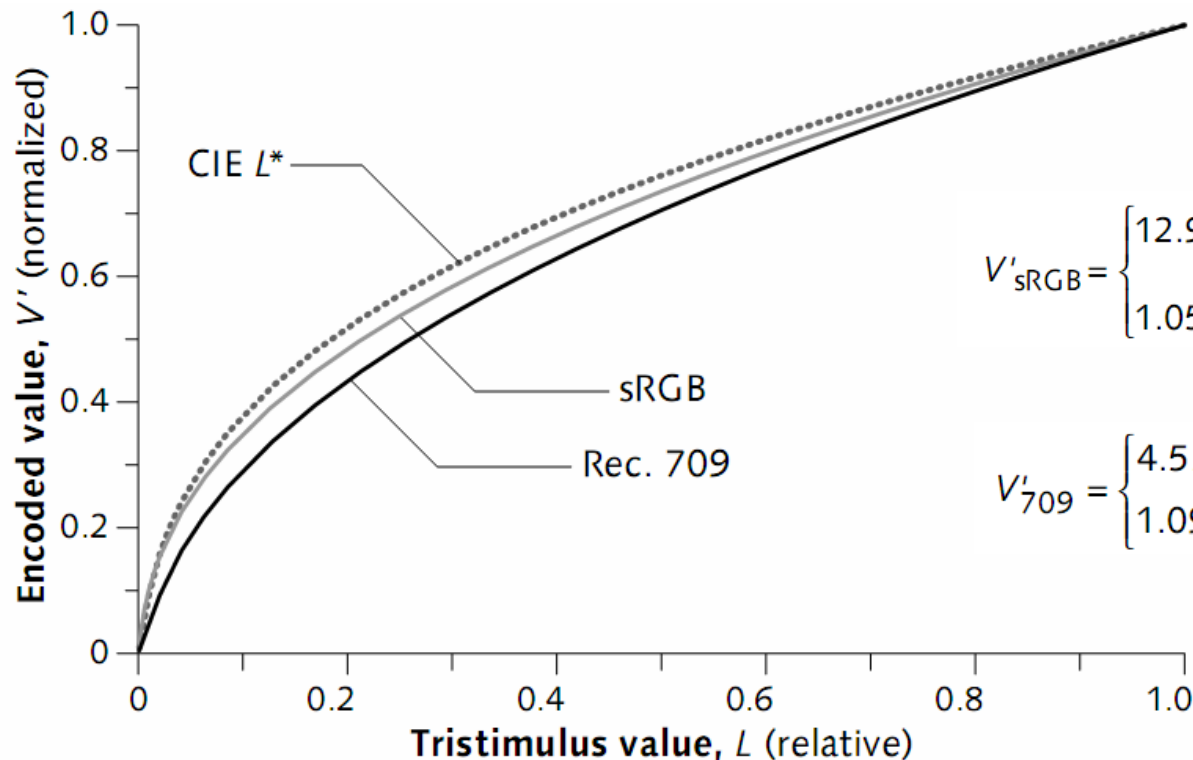


Note

- “white” object defined by its reflectance not by what it looks like
- Color space aims to “reproduce” equivalent color (spectrum) for human with different displays and illuminants
 - D65 white color looks the same with D32 illuminant
- White balance aims to “anticipate” human perception for white objects
 - D65 white color will look D32 white with D32 illuminant
 - Now, mostly used to create pleasing color

Electro-Optical Transfer Function (EOTF)

- Transform linear light to non-linear voltage
 - Linear RGB to non-linear R'G'B'



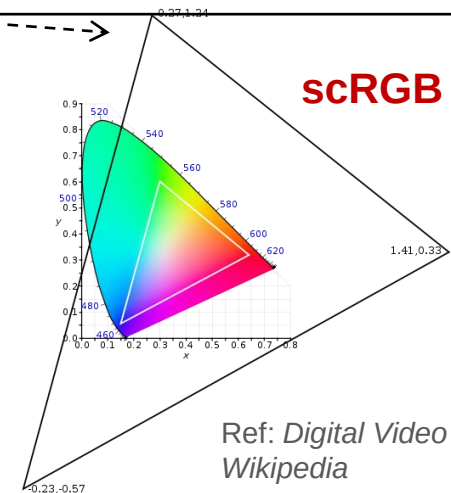
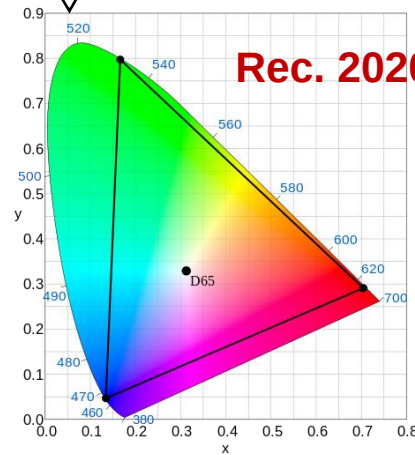
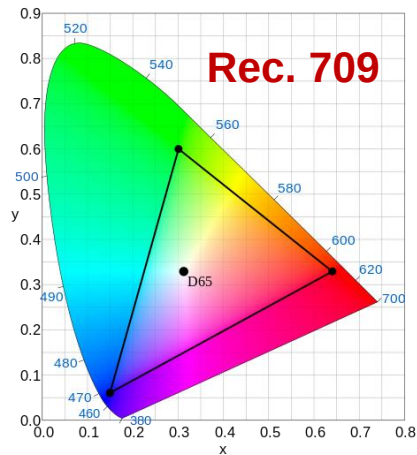
$$V'_{\text{sRGB}} = \begin{cases} 12.92L; & 0 \leq L \leq 0.0031308 \\ 1.055L^{\left(\frac{1}{2.4}\right)} - 0.055; & 0.0031308 < L \leq 1 \end{cases}$$

$$V'_{709} = \begin{cases} 4.5L; & 0 \leq L < 0.018 \\ 1.099L^{0.45} - 0.099; & 0.018 \leq L \leq 1 \end{cases}$$

Examples of Color Specification



Specification	Purpose	RGB Primaries (Gamut)	Electro-Optical Transfer Function (EOTF)	Luma/ Chroma Matrix	Digitization
sRGB	Computer	Rec. 709	sRGB	-	-
Rec. 601	SDTV	Rec. 709 w/ $x_g=0.29$	Rec. 709	Rec. 601	8-bit not full range
Rec. 709	HDTV	Rec. 709	Rec. 709	$\begin{bmatrix} 0.2126 & 0.7152 & 0.0722 \\ -0.1146 & -0.3854 & 0.5 \\ 0.5 & -0.4542 & -0.0458 \end{bmatrix}$	8-bit not full range
Rec. 2020	UHDTV	~Rec. 709	~Rec. 709	Y' : $[0.2627 \ 0.6780 \ 0.0593]$	10/12-bit not full range
scRGB	Device-indep.	Rec. 709 RGB: -2~2	Linear ($\gamma=1$)	-	16-bit -2 ¹⁵ ~2 ¹⁵ -1



Ref: Digital Video and HDTV,
Wikipedia

High Dynamic Range Video

- HDR10 vs Dolby Vision
 - EOTF: SMPTE ST 2084
 - RGB primaries: Rec. 2020
 - Bit depth: 10-bit for HDR10; up to 12-bit for Dolby Vision



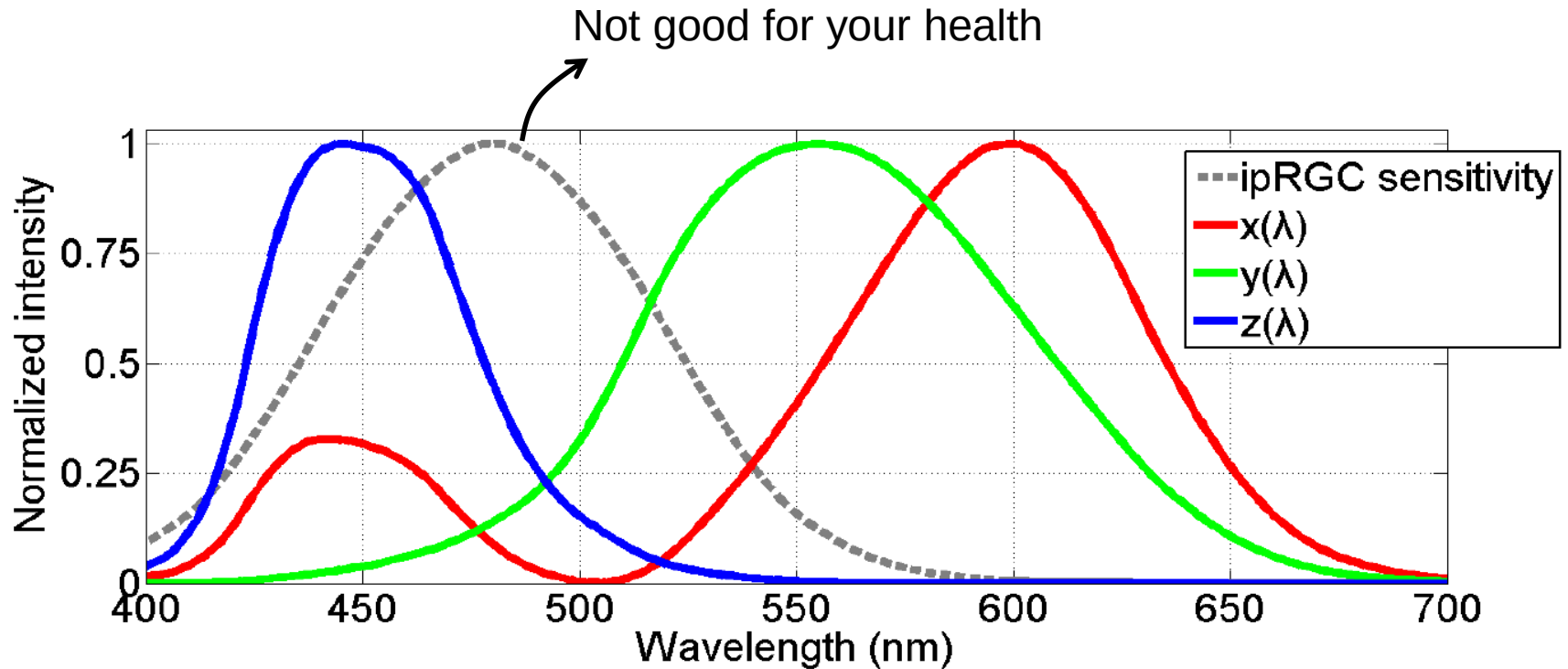
	HDR10	HDR10+	Dolby Vision
Peak Brightness	1k-4k Nits	1k-4k Nits	10k Nits
Bit Depth	10 Bit	10Bit	12 Bit
Resolution	4k	8k	8k
Content Availability	Great	Good	Good
HDR Type	Static HDR	Dynamic HDR	Dynamic HDR
Cost	\$	\$\$	\$\$\$
Platform Availability	Almost every Platform supports it	Netflix, Vudu, and Apple TV 4k isn't supported	Almost Every Platform with exception of NVIDIA Shield

Ref: https://en.wikipedia.org/wiki/High-dynamic-range_video (as of Jan 2021)



Appendix

Blocking Harmful Blue Light while Preserving Image Color Appearance



Is it possible at all to remove the harmful blue light without affecting color fidelity?

[K.-T. Shih et. al., SIGGRAPH Asia, 2016.]

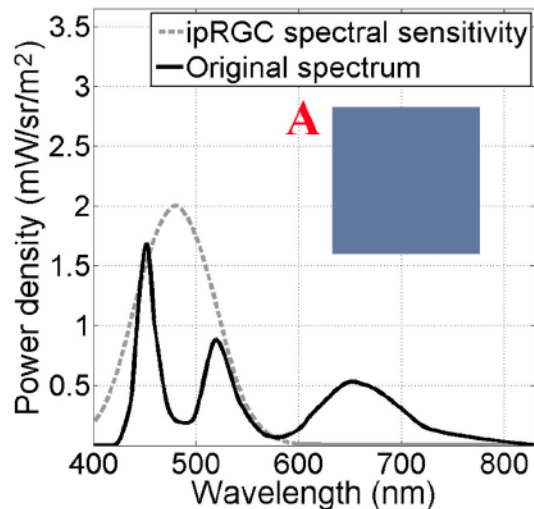
Blocking Harmful Blue Light while Preserving Image Color Appearance



(a) Original image

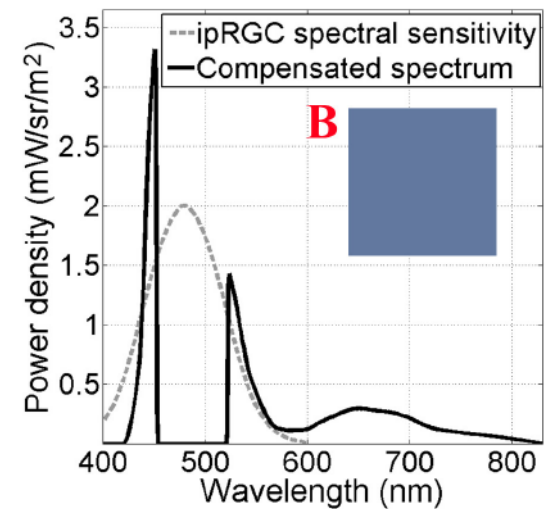
(b) Distorted image

(c) Our result



Use common
blue light blocker

Use proposed block
and optimization
algorithm



Blocking Harmful Blue Light while Preserving Image Color Appearance

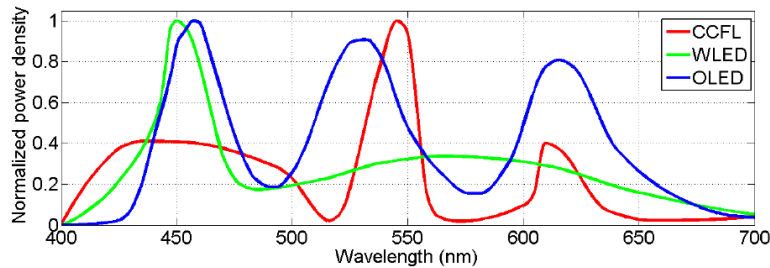


Figure. 3: The emission spectra of a cold cathode fluorescent lamp (CCFL), a white light-emitting diode (WLED), and an organic light-emitting diode (OLED).

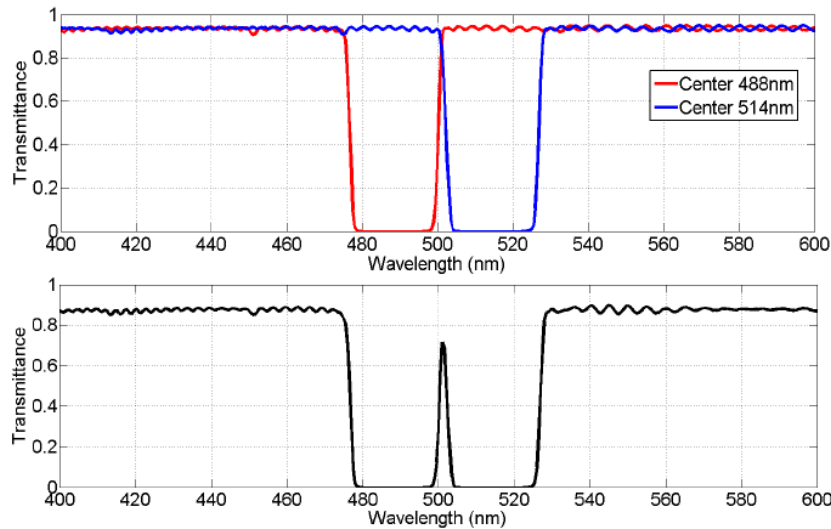


Figure. 13: (a) The spectral transmittance profile of the two notch filters we adopt in the prototype. (b) The spectral transmittance profile of the light blocker constructed by concatenating the two notch filters.

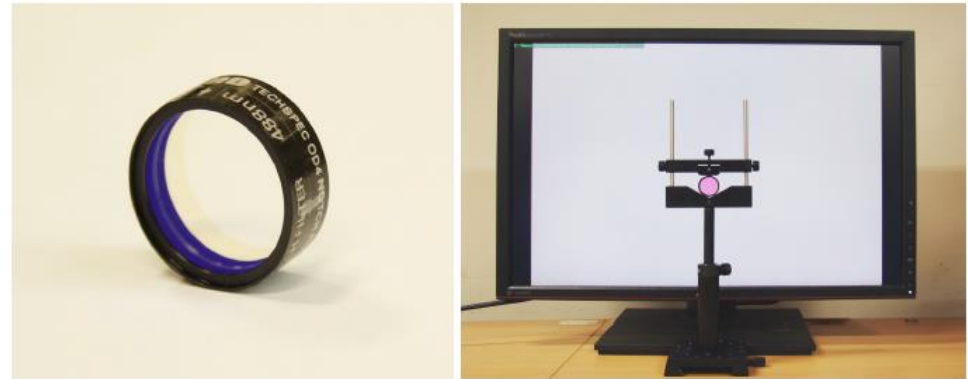
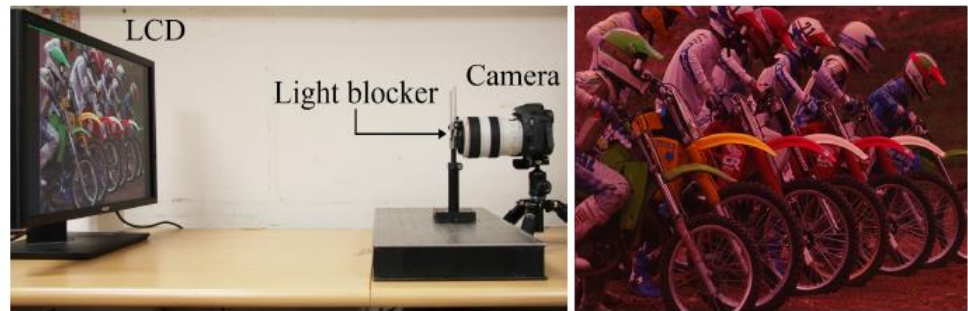


Figure. 14: (Left) The light blocker consisting of two cascaded optical notch filters. (Right) The light blocker is mounted on a holder and placed in front of an LCD.



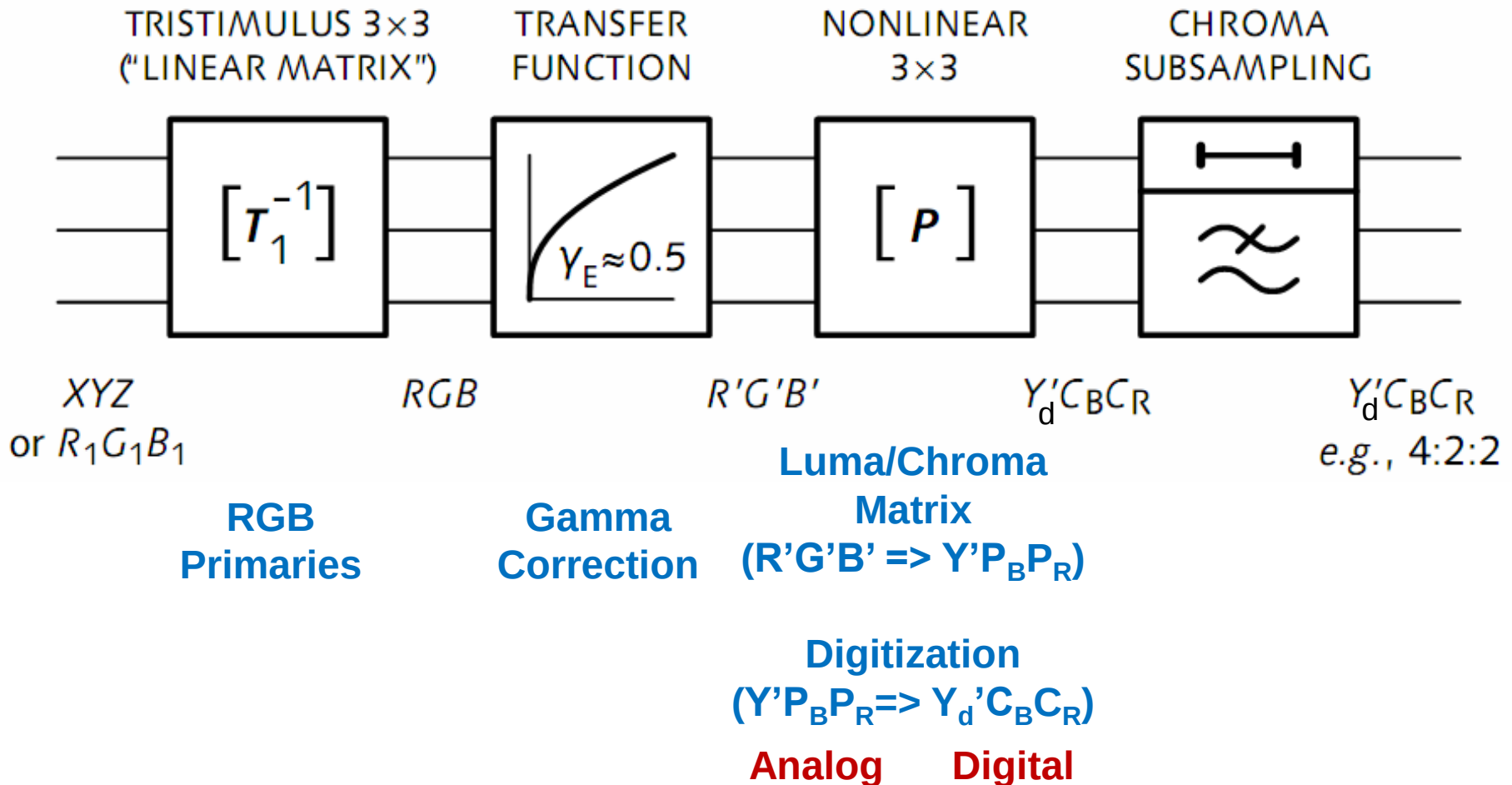
Blocking Harmful Blue Light while



Figure. 16: The original, distorted, and compensated images displayed on our prototype. All images are captured with a digital camera.

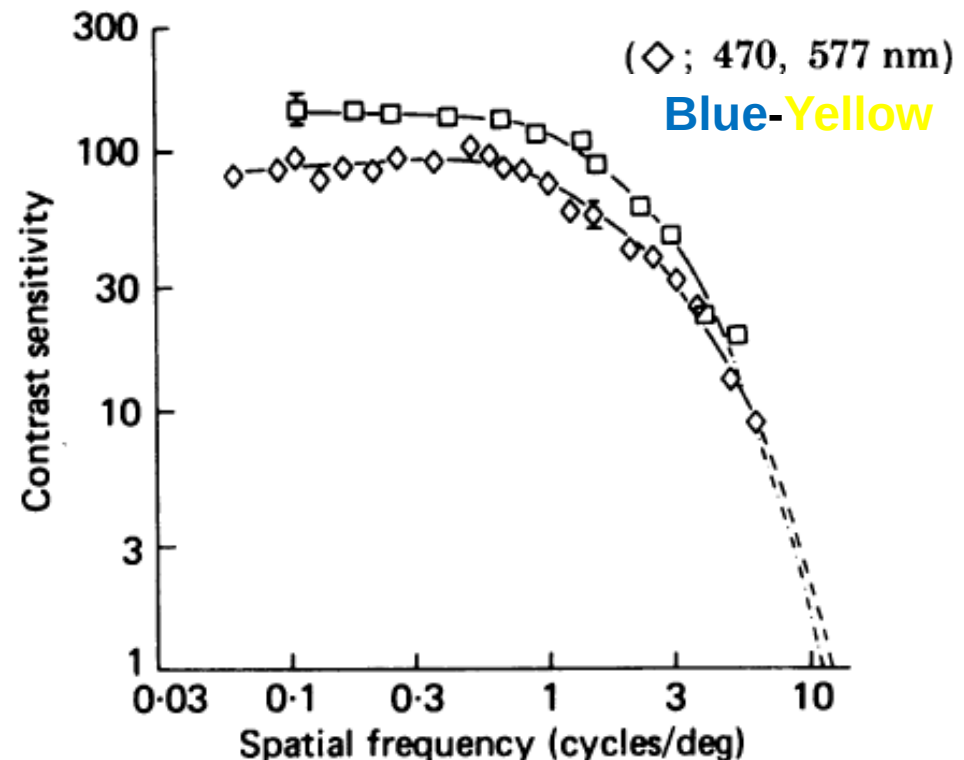
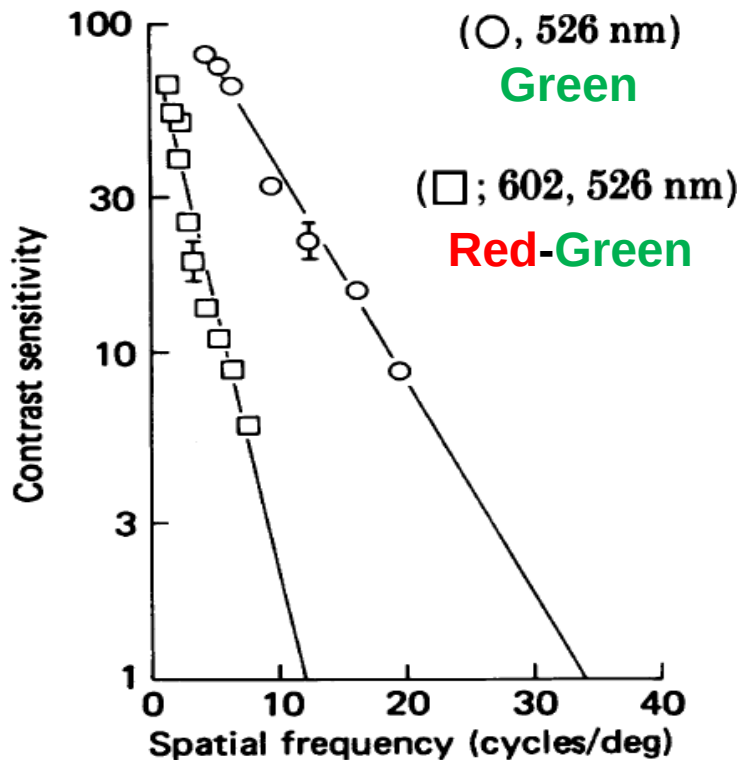


Color Specification for Digital Video Coding



Opponent Color Contrast Sensitivity

- Higher for White-Black (luminance $\approx$ green)
- Lower for Red-Green, Blue-Yellow (color difference)



Ref: "THE CONTRAST SENSITIVITY OF HUMAN COLOUR VISION TO RED-GREEN AND BLUE-YELLOW CHROMATIC GRATINGS", *J. Physiol.*, 1985.

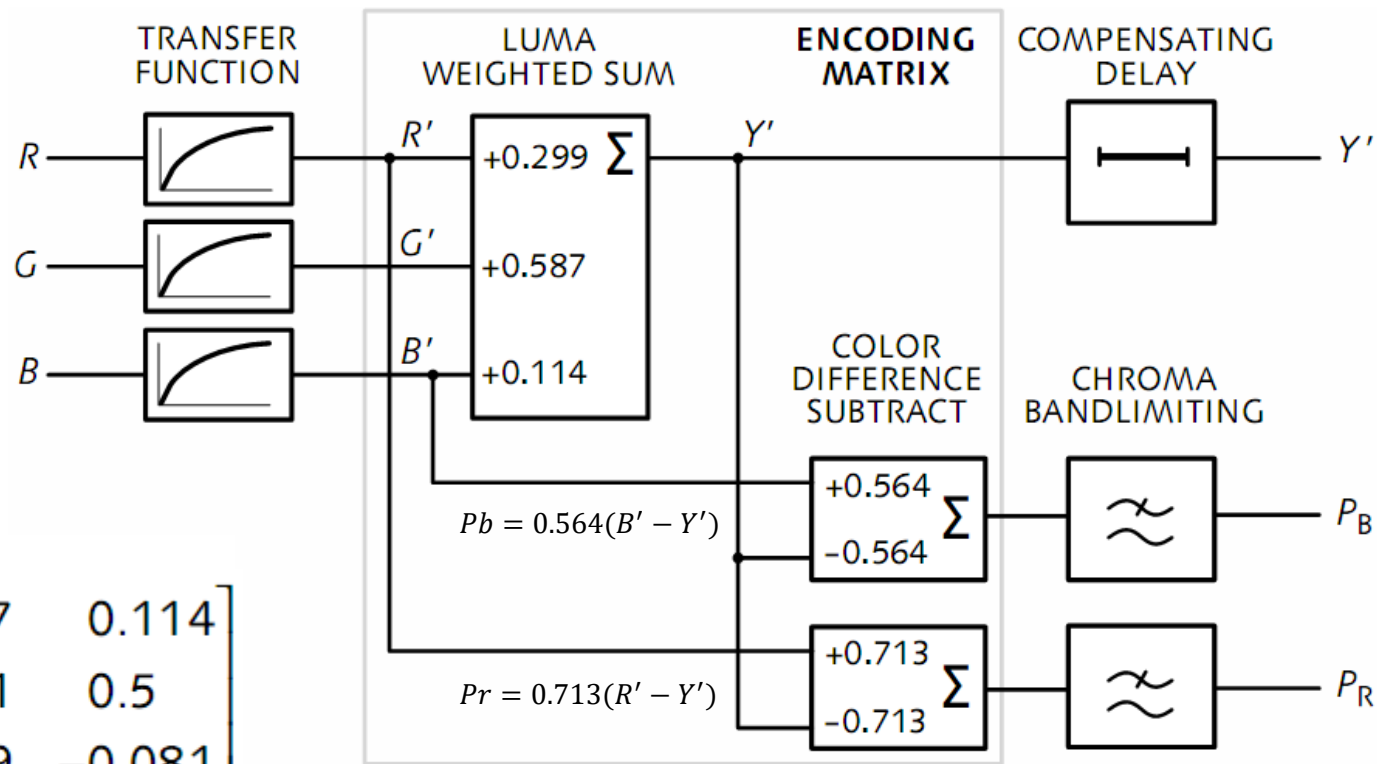
Luma and Chroma (non-linear)

- Luma Y' approximates luminance Y
- Chroma: color difference coding

Rec. 601

$$\begin{pmatrix} Y' \\ Pb \\ Pr \end{pmatrix} = P \begin{pmatrix} R' \\ G' \\ B' \end{pmatrix}$$

$$P = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.081 \end{bmatrix}$$



Ref: Digital Video and HDTV



Digitization

- Two types of common 8-bit coding

- Not full range

- $Y'_d = (219Y') + 16, C_L = (224P_L + 128)$ [L = B or R]

- [If $Y': 0 \sim 1, P_{L=B,R}: -0.5 \sim 0.5 \Rightarrow Y'_d: 16 \sim 235, C_{L=B,R}: 16 \sim 240$]

- Video full range

Avoid over- and under-shooting in analog TV

- $Y'_d = (255Y'), C_L = (255P_L + 128)$ [L = B or R]

- [If $Y': 0 \sim 1, P_{L=B,R}: -0.5 \sim 0.5 \Rightarrow Y'_d: 0 \sim 255, C_{L=B,R}: 0 \sim 255$]

Note: Names of chroma become a mess.

$Y P_B P_R$, $Y I Q$, $Y U V$ represent different analog chroma signals, but often misused.

Chroma Subsampling

- 4:4:4 is without subsampling

Since human less sensitive to chroma, usually subsampled to reduce data rate



4:2:2
progressive



4:2:0
MPEG-2
Frame
picture
(progressive)



4:2:0
JPEG/JFIF
(progressive)